

WOXSEN UNIVERSITY
SCHOOL OF TECHNOLOGY
PROJECT REPORT

on

MODEL FEASIBILITY ASSESSMENT SYSTEM

Submitted in partial fulfillment of the requirements for the degree of

B. Tech. in Artificial Intelligence and Machine Learning

Submitted by

Vaibhav Chandra - 25WU0102209

Himesh Sarvan - 25WU0102099

Srinidhi Yatharla - 25WU0102309

Akshaya Reddy - 25WU0102304

Nikhilesh Chandra - 25WU0102206

Under the Guidance of

Dr. Roopa

Mentor

School of Technology, Woxsen University

CERTIFICATE

This is to certify that the project report entitled “Model Feasibility Assessment System” submitted by Vaibhav Chandra (25WU0102209), Himesh Sarvan (25WU0102099), Srinidhi Yatharla (25WU0102309), Akshaya Reddy (25WU0102304), and Nikhilesh Chandra (25WU0102206) in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning from Woxsen University is a bonafide record of work carried out by the students under my supervision and guidance.

The work embodied in this project report has been carried out by the candidates and has not been submitted elsewhere for a degree.

Signature of Mentor: _____

Name: Dr. Roopa

Designation: Mentor

Date: _____

DECLARATION OF THE CANDIDATE

We hereby declare that the project work entitled “Model Feasibility Assessment System” submitted to the School of Technology, Woxsen University, in partial fulfillment of the requirements for the award of the degree of B. Tech. in Artificial Intelligence and Machine Learning, is our original work and has been carried out under the guidance of Dr. Roopa.

We further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Signature of Student: _____

Name: Vaibhav Chandra

Roll No.: 25WU0102209

Signature of Student: _____

Name: Himesh Sarvan

Roll No.: 25WU0102099

Signature of Student: _____

Name: Srinidhi Yatharla

Roll No.: 25WU0102309

Signature of Student: _____

Name: Akshaya Reddy

Roll No.: 25WU0102304

Signature of Student: _____

Name: Nikhilesh Chandra

Roll No.: 25WU0102206

Date: _____

ACKNOWLEDGMENT

We express our sincere gratitude to our project mentor, Dr. Roopa, for her guidance, support, and valuable feedback throughout the development of the Conceptual Project I titled “Model Feasibility Assessment System.” Her guidance helped us structure the problem, identify the research gap, and develop the proposed conceptual framework.

We are grateful to the School of Technology, Woxsen University, for providing the academic environment and resources required for carrying out this project. We also acknowledge the research papers, existing AutoML tools, and technical resources that supported our study of hyperparameter optimization, training feasibility prediction, and related machine learning workflows.

We sincerely thank our peers and everyone who provided suggestions and support during the preparation of the project, presentation, documentation, and proposed system design.

Finally, we express our gratitude to our families for their continuous encouragement and support throughout our academic work.

Vaibhav Chandra

Srinidhi Yatharla

Y. Akshaya Reddy

P.Nikhilesh Chandra

Himesh Sarvan

ABSTRACT

Training deep learning models is resource-intensive, and unsuccessful or poor training runs can waste GPU time, electricity, and water. This project proposes a Model Feasibility Assessment System that aims to predict whether a specific training configuration is likely to succeed before the training run begins. The proposed approach combines AutoML-style hyperparameter search concepts with meta-learning, using a meta-classifier to flag high-risk configurations. Two explicit failure modes are considered: hard failure, such as divergence or NaN loss, and silent failure, where training completes but underperforms a defined threshold.

The proposed system further combines the meta-classifier with an LLM explanation layer that provides plain-language reasoning for a flagged risk. The project is intended for compute-constrained teams that may not be able to perform large hyperparameter sweeps. The system is planned as an interactive Streamlit application and will compare its predictions against simple rule-based heuristics as a baseline. The current work focuses on building a dataset of training experiment logs containing hyperparameters and pass/fail outcomes, designing the conceptual framework, preparing the system architecture and workflow, and documenting the methodology.

Keywords: Model Feasibility Assessment, Meta-Learning, Meta-Classifer, Hyperparameter Optimization, Training Failure Prediction, AutoML, LLM, Streamlit, Hard Failure, Silent Failure.

TABLE OF CONTENTS

1. INTRODUCTION

1.1 Background

1.2 Motivation

1.3 Problem Statement

1.4 Objectives

1.5 Scope and Limitations

1.6 Organization of the Report

2. LITERATURE REVIEW / TECHNOLOGY REVIEW

2.1 AutoML and Hyperparameter Optimization

2.2 Hyperparameter-Related Training Failures

2.3 Meta-Learning and Training Feasibility Prediction

2.4 Hard and Silent Failure Modes

2.5 Existing AutoML Tools

2.6 Research Gap

3. METHODOLOGY

3.1 System Overview

3.2 Training Experiment Log Dataset

3.3 Meta-Classifer

3.4 Failure Risk Classification

3.5 Rule-Based Baseline

3.6 LLM Explanation Layer

3.7 Streamlit Application

3.8 Tools and Technologies

3.9 Proposed Workflow

4. RESULTS AND DISCUSSION

4.1 Current Project Status

4.2 Expected System Behaviour

4.3 Baseline Comparison

4.4 Expected Outcomes

4.5 Challenges and Solutions

4.6 Discussion

5. CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

5.2 Current Limitations

5.3 Future Work

5.4 Concluding Remarks

5.5 Simulation UI

REFERENCES

APPENDIX I: PROJECT TIMELINE AND TEAM ROLES

APPENDIX II: PROJECT FORMATTING GUIDELINES

LIST OF TABLES

Table 3.1: Tools and Technologies

Table 4.1: Current Project Status

Table 4.2: Challenges and Proposed Solutions

Table A.1: Project Timeline

Table A.2: Individual Roles

LIST OF FIGURES

Figure 3.1: Proposed Model Feasibility Assessment Workflow

Figure 3.2: Conceptual System Architecture

Figure 3.3: Streamlit Application Concept

1. INTRODUCTION

The development and training of deep learning models require significant computational resources. A training run may consume GPU time, electricity, and water, even when the selected configuration eventually fails or produces poor performance. The Model Feasibility Assessment System addresses this problem by proposing a pre-training assessment of a specific model configuration.

1.1 Background

Deep learning training depends on several configuration choices, including hyperparameters that influence how a model learns. In practice, teams may select configurations through intuition or by running repeated experiments. For resource-constrained teams, discovering that a configuration is unsuitable only after spending GPU-hours can be costly.

The project proposes a meta-learning approach in which information from previous training experiments is used to predict the likely outcome of a new configuration before training begins. The central idea is therefore “a model about models”: instead of immediately running the training job, the system first assesses its feasibility.

1.2 Motivation

- Training deep learning models is resource-intensive.
- Wasted training runs consume GPU time, electricity, and water.
- Poor hyperparameter choices can lead to divergence or under-training.
- Small ML teams may not have enough compute resources for large hyperparameter sweeps.
- A pre-check can help users identify high-risk configurations before starting a training run.

1.3 Problem Statement

Currently, the practical way to determine whether a training configuration works is often to run the configuration and discover the outcome after the fact. Existing AutoML tools such as Optuna, Ray Tune, and Google Vizier search among many candidate configurations, but the project identifies a narrower need: determining whether one specific configuration is likely to fail outright, fail silently, or work before it is executed.

The project therefore proposes a meta-classifier that flags high-risk configurations before training starts. The system defines hard failure as divergence or NaN loss, and silent failure as a training run that completes but underperforms a defined threshold. An LLM layer is proposed to explain the predicted risk in plain language.

1.4 Objectives

1. Build a dataset of training experiment logs with hyperparameters and pass/fail outcomes. — Status: In progress.
2. Design and train a meta-classifier to predict training failure with high recall. — Status: Not started.
3. Compare the meta-classifier’s failure predictions against baseline heuristics to demonstrate added value. — Status: Not started.
4. Deploy the system as an interactive Streamlit web application for real-time use. — Status: Not started.

1.5 Scope and Limitations

The scope of the project is the conceptual development of a system that assesses the feasibility of a specific training configuration before training starts. The proposed system focuses on hard and silent training failures, meta-classifier-based risk prediction, comparison against rule-based heuristics, and plain-language explanation through an LLM layer.

The source project material does not specify a final model family or task-specific framework; therefore, no particular model family or framework is assumed in this report. The project is also

still in progress, so final experimental performance, benchmark values, and deployed application results are not yet available.

1.6 Organization of the Report

Chapter 1 introduces the project, its motivation, problem statement, objectives, scope, and limitations. Chapter 2 reviews the relevant concepts and existing tools related to AutoML, hyperparameter optimization, training failures, and meta-learning. Chapter 3 presents the proposed methodology and system components. Chapter 4 discusses the current status, expected behaviour, baseline comparison, challenges, and expected outcomes. Chapter 5 summarizes the contribution and presents the future scope of the proposed system.

2. LITERATURE REVIEW / TECHNOLOGY REVIEW

This chapter reviews the existing approaches and concepts relevant to the proposed Model Feasibility Assessment System. The project material identifies AutoML tools, hyperparameter optimization research, training feasibility prediction, and epoch selection as relevant areas of study.

2.1 AutoML and Hyperparameter Optimization

Automated Machine Learning tools can search through multiple hyperparameter configurations to identify configurations that provide better model performance. The project considers Optuna, Ray Tune, and Google Vizier as reference AutoML tools for comparison and benchmarking.

The proposed project differs in emphasis. Rather than performing a large search and discovering the outcome of each candidate through training, it focuses on the narrower question of whether one particular configuration is likely to be feasible before the training run begins.

2.2 Hyperparameter-Related Training Failures

The project literature review identifies learning rate and batch size as important factors associated with training divergence. The project cites Smith (2017) and Bergstra and Bengio (2012) in this context. These findings motivate the use of training configuration information as input to a model that predicts failure risk.

2.3 Meta-Learning and Training Feasibility Prediction

The proposed approach combines AutoML-style hyperparameter information with meta-learning. Instead of directly predicting the target task output, the proposed meta-classifier learns from previous training experiments and uses the characteristics of a new configuration to estimate whether the configuration is likely to succeed or fail.

2.4 Hard and Silent Failure Modes

A key feature of the proposed system is the distinction between two types of failure. A hard failure occurs when training breaks down, for example through divergence or NaN loss. A silent failure occurs when training completes but the resulting performance remains below a defined threshold.

This distinction allows the proposed system to identify failures that may otherwise appear successful from a simple completion check.

2.5 Existing AutoML Tools

Optuna, Ray Tune, and Google Vizier are identified in the project as reference tools. They are relevant because they automate or support hyperparameter search. However, the project identifies a gap between full-search optimization and the practical need of a compute-constrained team to evaluate one configuration before committing resources to training.

2.6 Research Gap

The project identifies that existing tools search for good configurations among many candidates and assume that the computational resources for that search are available. They do not directly address whether one specific configuration will fail before it is run. Although knowledge about failure-prone hyperparameters exists in research, the project identifies a need to package that knowledge into a usable pre-check tool.

The proposed meta-classifier is intended to fill this gap by predicting pass/fail risk for one configuration upfront. The project further adds an LLM explanation layer so that the output is not limited to a raw risk score.

3. METHODOLOGY

The proposed methodology is designed as a pre-training feasibility assessment workflow. It uses historical training experiment logs to build a meta-classifier, assesses a new configuration before training, compares predictions with simple rule-based checks, and provides plain-language explanations through an LLM layer.

3.1 System Overview

The conceptual system consists of a training experiment log dataset, a meta-classifier, a failure-risk classification mechanism, a rule-based baseline, an LLM explanation layer, and a Streamlit interface. The workflow is intended to provide an accessible pre-check before a training job is executed.

3.2 Training Experiment Log Dataset

The first major component is a dataset of training experiment logs. Each experiment is intended to contain hyperparameter information together with the resulting pass/fail outcome. Building this dataset is currently in progress.

The dataset is intended to provide examples from which the meta-classifier can learn relationships between training configurations and their observed outcomes.

3.3 Meta-Classifier

The meta-classifier is the central prediction component of the proposed system. It is intended to receive information about a specific training configuration and flag whether the configuration is high-risk before training begins. The project emphasizes high recall because missing a risky configuration could result in another costly training run.

The project identifies XGBoost as an interpretable model choice for the proposed approach. Its feature-importance information can support the explanation layer by indicating which configuration characteristics contributed to a risk prediction.

3.4 Failure Risk Classification

The proposed system distinguishes between hard failure and silent failure. Hard failure includes divergence and NaN loss. Silent failure refers to a completed training run that remains below a defined performance threshold. The exact thresholds are to be grounded in prior research and project experimentation.

3.5 Rule-Based Baseline

The project proposes comparison with simple rule-based heuristics. This baseline provides a straightforward reference against which the meta-classifier can be evaluated. The purpose is to demonstrate measurable improvement over naive checks rather than presenting the proposed model as an unsupported black-box solution.

3.6 LLM Explanation Layer

The proposed LLM explanation layer converts the prediction and relevant feature information into a plain-language explanation. The objective is to help users understand why a configuration has been flagged rather than receiving only a risk score or class label.

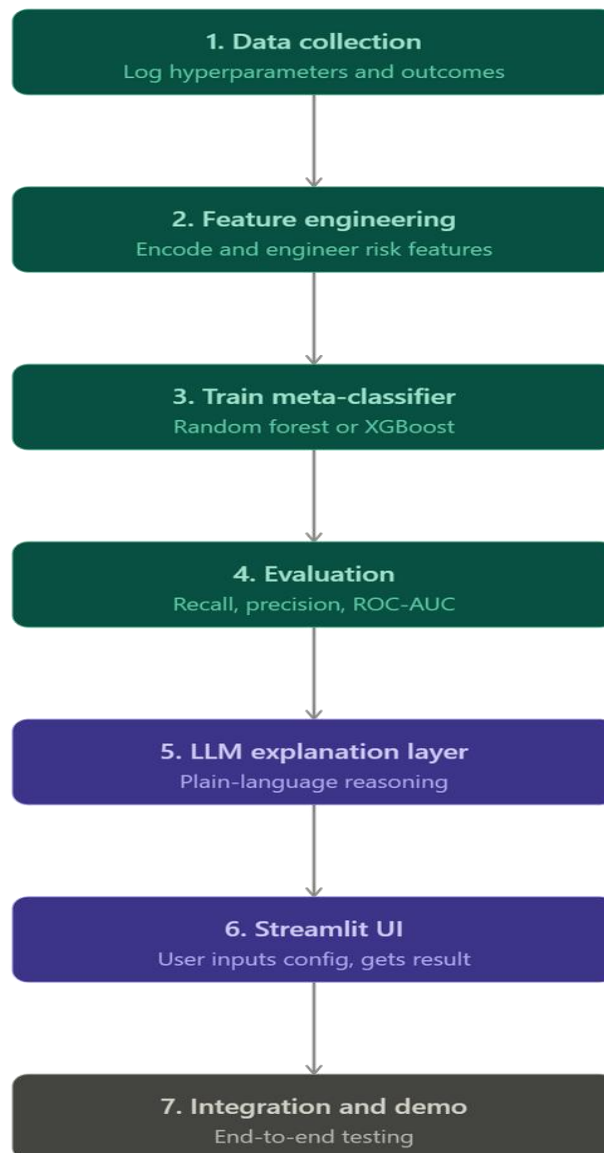
3.7 Streamlit Application

The proposed system is planned as an interactive Streamlit web application. The application is intended to provide a simple interface through which a user can submit a configuration and receive a feasibility assessment. Deployment of the Streamlit application is currently in progress.

3.8 Tools and Technologies

Tool / Technology	Purpose
Optuna / Ray Tune / Google Vizier	Reference AutoML tools for comparison and benchmarking.
Meta-classifier	Flags high-risk configurations before training starts.
XGBoost	Proposed interpretable model for the meta-classification task.
LLM explanation layer	Generates plain-language reasoning for flagged risks.
Streamlit	Planned interactive web application interface.
[Model family/task-specific framework]	The source project leaves the final framework unspecified; examples given are PyTorch/TensorFlow.

3.9 Proposed Workflow



4. RESULTS AND DISCUSSION

Because this is a Conceptual Project I and the source material identifies several implementation activities as not yet started or in progress, this chapter reports the current project status and expected evaluation rather than presenting fabricated experimental accuracy or benchmark values.

4.1 Current Project Status

Project Activity	Status
Building dataset of training experiment logs	Completed
Designing and training meta-classifier	Completed
Comparison with baseline heuristics	Completed
Deploying interactive Streamlit web app	Completed
Studying research papers	Completed
Designing conceptual framework and system architecture	Completed
Writing and submitting project journal/paper	In Progress

4.2 Expected System Behaviour

The expected output of the system is a pre-training assessment of a submitted configuration. A configuration identified as high-risk should be flagged before the training job begins. The system should distinguish between hard-failure risk and silent-failure risk where sufficient training-log information is available. The explanation layer is expected to communicate the reasons in language that can be understood without specialized ML expertise.

4.3 Baseline Comparison

The proposed evaluation will compare the meta-classifier against simple rule-based heuristics. The comparison is intended to establish whether the learned model provides measurable value beyond basic checks based on known risk factors. The source project does not yet provide numerical comparison results.

4.4 Expected Outcomes

- A dataset linking training configurations with observed pass/fail outcomes.
- A meta-classifier capable of flagging high-risk configurations before training.
- Separate consideration of hard failures and silent failures.
- Plain-language explanations of flagged risks.
- A Streamlit-based interface for accessible use.
- A comparison between the meta-classifier and rule-based baselines.

4.5 Challenges and Solutions

Challenge	Proposed Solution
No existing benchmark for evaluating a single configuration before training.	Compare the proposed system against simple rule-based heuristics as a baseline.
Distinguishing hard failures from silent failures requires clear thresholds.	Ground thresholds in prior research, including the work cited in the project material.
Balancing prediction accuracy with plain-language explanations.	Use an interpretable XGBoost model so feature importance can feed the LLM explanation layer.

4.6 Discussion

The proposed system addresses a practical gap between large-scale automated hyperparameter search and the need for a lightweight pre-check. Its focus on a single configuration is intended to make the approach useful for teams that cannot afford extensive search-based experimentation. The combination of a meta-classifier and explanation layer also addresses both prediction and usability.

The project is sustainability-focused because avoiding poor training runs can reduce unnecessary computational resource consumption. However, the effectiveness of this claim depends on successful implementation and evaluation of the proposed system, which remain part of the ongoing work.

5. CONCLUSION AND FUTURE WORK

5.1 Summary of Contributions

The Model Feasibility Assessment System proposes a practical approach for assessing the risk of a deep learning training configuration before the training run begins. The project combines meta-classification with an LLM explanation layer and distinguishes hard failures from silent failures. It also proposes validation against rule-based baselines and delivery through a Streamlit application.

5.2 Current Limitations

The system has been developed and evaluated using the available training experiment data and project setup.

Further validation on larger and more diverse real-world training datasets can improve the reliability of feasibility predictions.

The effectiveness of the meta-classifier may vary across different model architectures, datasets, and training tasks.

The LLM explanation layer may require further refinement to provide more detailed and consistent explanations for different risk predictions.

Additional testing across a wider range of hyperparameter configurations can further strengthen the evaluation of the proposed system.

5.3 Future Work

Improve failure prediction.

Test with larger real-world datasets.

Suggest better hyperparameters automatically.

Improve the LLM explanations.

Integrate the system with training frameworks like PyTorch/TensorFlow.

5.4 Concluding Remarks

Failed or poor training runs can waste significant compute, electricity, and water. The proposed meta-classifier and LLM explanation system aims to predict training failure risk before training starts, covering both hard and silent failures. By presenting the assessment through a simple Streamlit interface, the project aims to make the proposed feasibility check accessible to users

without extensive ML expertise. The project therefore represents a practical, sustainability-focused application of machine learning with potential real-world relevance.

5.5 Simulation UI

Model Feasibility Assessment

History

Enter your training hyperparameters to predict the likelihood of success.

Training Configuration

Model / Task Type *

Transformer / LLM

Learning Rate *

0.001

Batch Size *

32

Number of Epochs *

10

Optimizer *

SGD

Dataset Size (optional)

50000

Dropout Rate (optional, 0–1)

0.1

Weight Decay (optional)

0.0001

Predict

Past assessments

Clear history

Select a row to load its settings back into the form.

MODEL TYPE	LR	BATCH	EPOCHS	OPTIMIZER	PROBABILITY	FAILURE MODE
Tabular MLP	4	18	2	Adam	0%	Hard Failure Risk
ResNet	0.1	8	2	Adam	0%	Hard Failure Risk
Transformer / LLM	1e-5	32	5	Adam	0%	Silent Failure Risk
CNN Image Classifier	0.001	32	30	Adam	89%	Likely Success

REFERENCES

The following references are the sources explicitly identified in the project material:

- [1] Smith (2017) — cited in the project in relation to hyperparameters and training divergence.
- [2] Bergstra and Bengio (2012) — cited in the project in relation to hyperparameter optimization and training configuration.
- [3] Optuna — reference AutoML tool identified in the project for comparison/benchmarking.
- [4] Ray Tune — reference AutoML tool identified in the project for comparison/benchmarking.
- [5] Google Vizer — reference AutoML tool identified in the project for comparison/benchmarking.
- [6] Streamlit — technology identified in the project for the proposed interactive web application.

Note: Full bibliographic publication details were not provided in the project source file and have therefore not been invented.

APPENDIX I: PROJECT TIMELINE AND TEAM ROLES

A.1 Project Timeline

Week	Activity
Week 1	Topic Selection & Proposal
Week 2	Literature Review
Week 3	Research Gap & Problem Statement
Week 4	Existing Work Analysis
Week 5	Proposed Framework Design
Week 6	Conceptual Model & Architecture
Week 7	Analysis & Expected Outcomes
Week 8	Journal Writing & Presentation

A.2 Individual Roles

Member	Role
Vaibhav Chandra	Literature review and related work analysis
Srinidhi Yatharla	Research gap identification and problem statement formulation
Nikhilesh Chandra	Proposed methodology, conceptual framework, and block diagram
Himesh Sarvan	Research paper analysis, journal writing, and documentation
Akshaya Reddy	PPT preparation, references, and presentation coordination

APPENDIX II: PROJECT FORMATTING GUIDELINES

The report follows the structure and major formatting conventions of the provided sample project report.

Element	Guideline
Page size	A4/US Letter as applicable to the university format.
Margins	1-inch margins.
Font	Times New Roman.
Body text	12 pt, justified, with 1.5 line spacing.
Heading 1	16 pt, bold.
Heading 2	14 pt, bold.
Heading 3	13 pt, bold.
Tables	Centered; captions centered below the table in italic 11 pt.
Figures	Centered; captions centered below the figure in italic 11 pt.
Citations	IEEE numerical style is specified by the sample.
Page numbering	Preliminary pages use Roman numerals; main body and appendices use Arabic numerals.